Answers to Theoretical Exercises

Part I: Basic Concepts and Quantities

1. **Human life course.** The left panel shows the probability (solid line) and the hazard (dashed line). The probability curve shows that the most likely age of death is around 80 years. The hazard increases rapidly after 60 years: the older you are, the more likely you are to die within the coming year. The right panel shows the survival curve. The slope of the survival curve corresponds to the probability: the steeper the curve, the more individuals die at that age.

2. **Censoring due to discharge.** We want to compare the distribution of time to side effects between both hospitals. Discharge is not a competing risk, because it does not prevent the "side effects" event to occur. Since there is only a single event type involved, it does not make sense to call it a marginal distribution. The comparison is fair if we can assume that side effects are equally likely to occur in individuals who stay in the hospital as in individuals that have been discharged.

3. **Definition of cause-specific hazard.** The cause-specific hazard is commonly defined as the hazard in the *presence* of competing risks. The hazard in the absence of competing risks is usually called the marginal hazard, net hazard or simply the hazard.

   The statement has been copied from [4, p.1362]. Latouche *et al.*[3] replied:

   > Pintilie's work has the potential to further obscure the issues . . . Our main critique concerns the inaccurate assertion: "When modelling the cause specific hazard, one performs the analysis under the assumption that the competing risks do not exist".

4. **Definition of competing risk.** An event that precludes the occurrence of the event of interest certainly is a competing risk. If the event fundamentally alters the probability of the event of interest, it may be treated as a competing risk; it depends on the study question. Between birth and death, many event can happen that alter the death risk. But we ignore them if we are only interested in the overall mortality. Or if we want to quantify the natural history from HIV infection to AIDS, irrespective of what happens in-between, we may ignore the SI switch and we are in the classical setting of a single event type.

   Also, the definition seems to focus only on the situation that a single event type is of main interest.

Part II: Estimation: Multi-State Approach

1. **Ignore the competing event?** The Kaplan-Meier is a valid estimator of the marginal distribution if all censoring is non-informative. Removing individuals from the risk set before they experience the competing event removes the competing risk, but does not solve the problem of informative censoring.

2. **What if Kaplan-Meier and Aalen-Johansen are almost equal?** The statement seems to suggest that the Kaplan-Meier and the estimator of the CE-specific cumulative incidence are equal if censoring is non-informative. This is nonsense; they quantify different aspects. See the example with discharge and staphylococcus infection as competing risks. The reason that both curves are very similar is because there is little mortality due to other causes, at least during the first 20 years,

when most of the symptomatic cardiac events occur. Also note that on one hand it is said that death due to other causes may not be related to CE events, whereas on the other hand it is called "informative censoring". The statement was inspired by a similar one on breast cancer and ovarian cancer in Satagopan *et al.* [6, p.1234].

3. **Interpretation of gender difference.** The comparison does not tell us anything on the reason for the gender difference in relapse-specific cumulative incidence. The Kaplan-Meier is supposed to estimate the marginal distribution, which quantifies a different aspect than the cause-specific cumulative incidence. See the example with discharge and staphylococcus infection as competing risks. Moreover, it is only valid if DOC is noninformative for relapse. If DOC is informative for relapse, it does not quantify what would happen if DOC did not exist. The time-to-event data does not provide the information to determine the relation between DOC and relapse. For example, it could be that every person that died would have progressed on the next day. This would be the same as adding the cause-specific cumulative incidences of relapse and death. Hence, the most extreme scenario which cannot be excluded based on the data would be obtained by combining the two event types, or adding the two cause-specific cumulative incidence curves. Then, the estimated marginal cumulative incidence curve for the males will become almost similar to the one for the females.

   The reason that the Kaplan-Meier and the estimate of the cause-specific incidence are almost the same is because there is little mortality due to other causes, at least during the first 40 months, when most of the relapses occur.

   The Kaplan-Meier does give information on the cause-specific hazards (it has a one-to-one relation, even though the Kaplan-Meier itself may not be interpretable). We can conclude that females have a higher relapse-specific hazard than males and females have a lower DOC-specific hazard than males.

   This exercise was inspired by the PhD thesis of Porta-Bleda [5, p.32].

4. **Cause-specific hazard and independent censoring.** Statement I: By definition, a cause-specific model quantifies the effect of a covariable on an event *in the presence of* competing risks. The statement is correct insofar as the comparison between the combined end point and a specific event in the presence of competing risks is concerned.

   Statement II: A cause-specific Cox regression indeed treats competing events as censored. But competing risks are not ignored with respect to the interpretation of the estimates.

   The statement resembles the one in Exercise I.10.3. Of course, one can use another definition of cause-specific, making it correspond with marginal, but that will only generate more confusion.

   The statements were adapted from Kim [2, p.559 and p.565].

Part III: Estimation: Subdistribution Approach

1. **Log-rank test?** In competing risks settings, we do not want to compare Kaplan-Meier cumulative incidence curves. However, the standard log-rank test is a valid test for equality of cause-specific hazards. The statement is adapted from Kim [2, p.562].

Part IV: Study Questions

1. **Cause-specific and subdistribution as surrogate of marginal.** The cause-specific hazard is often interpeted as marginal hazard. Sometimes this happens because of plain ignorance, sometimes it is deliberately chosen as surrogate measure. However, we have seen that it is not always justified and marginal hazard ratios may be different (see the example of time to AIDS in IDUs).

   In general, marginal hazard and subdistribution hazard are different concepts. If the competing events are independent, regression on the subdistribution hazard has no relation to regression on the marginal hazard. They are equal only if individuals who experience the competing risk would never have experienced the event of interest, as is the case when such individuals are "cured" for the event of interest.

   The statement is an adapted version of one made in [7, p.460].

2. **From cause-specific to subdistribution.** We use the relation

   $$F_k(t) = \int_0^t \lambda(s)\overline{F}(s)ds$$

   $\overline{F}(s)$ can be written as $\exp\{-[\Lambda_1(s) + \Lambda_2(s)]\}$. Λ_1 is the same in both groups, but $\Lambda_2(s|I) > \Lambda_2(s|II)$. Therefore, $\overline{F}(s|I) < \overline{F}(s|II)$, which implies $F_1(t|I) < F_2(t|I)$. By the one-to-one relation between cause-specific cumulative incidence and subdistribution hazard, we derive that the subdistribution hazard for cause 1 is smaller in group I.

3. **Competing risks is Fine and Gray?** Since they address "potential bias", the purpose of their study seems to be the causal effect of β-blockers. Such an effect is better quantified by the cause-specific hazard. It may be that the lower subdistribution hazard for PCa-specific mortality in individuals that use β-blockers is completely explained by the fact that they have higher cardiac mortality. In the most extreme situation that they would die of prostate cancer shortly after their observed DOC, then the marginal distribution for PCa-specific mortality is approximated by the overall mortality, which is more or less equal. If individuals live shorter, prostate cancer is less likely to develop, but this does not necessarily imply a causal effect. A similar argument against their use of the Fine and Gray model was given in [1].

4. **Treatment failure under interventions.** There is no guarantee that it is a valid explanation. A model for the subdistribution hazard was used. Let's simplify the setting and assume that treatment failure and treatment interruption are the only two competing events. If the ones who interrupt treatment are less likely to fail, the most extreme scenario occurs when they will never fail. Then the marginal hazard would be the same as the subdistribution hazard and the marginal hazard ratio would be equal to the reported one.

   It could be a possible explanation if the observed hazard ratios were from a model for the cause-specific hazard. If the individuals that interrupt treatment are less likely to fail, they would remain in the risk set for a longer period if they had not interrupted. Since a higher percentage interrupts in the early starters, it concerns a larger fraction and thus has a bigger impact.

   We can also show it mathematically. We again consider the extreme situation that individuals that interrupt treatment would never fail. A larger interruption-specific hazard in early starters and equal failure-specific hazards in both groups implies a smaller subdistribution hazard for failure in the early starters. This property was shown in exercise IV.1. By equality, the marginal hazard is smaller as well in the early starters.

References

[1] K. Bhaskaran, B. Rachet, S. Evans, and L. Smeeth. Re: Helene Hartvedt Grytli, Morten Wang Fagerland, Sophie D. Fosså, Kristin Austlid Taskén. Association between use of β-blockers and prostate cancer-specific survival: a cohort study of 3561 prostate cancer patients with high-risk or metastatic disease. *European Urology*, 64(4):e86–87, 2013.

[2] Haesook T Kim. Cumulative incidence in competing risks data and competing risks regression analysis. *Clinical Cancer Research*, 13(2):559–565, 2007.

[3] A. Latouche, J. Beyersmann, and J. P. Fine. Comments on 'analysing and interpreting competing risk data'. *Statistics in Medicine*, 26(19):3676–9; author reply 3679–80, 2007.

[4] Melania Pintilie. Analysing and interpreting competing risk data. *Statistics in Medicine*, 26(6):1360–1367, 2007.

[5] Núria Porta-Bleda. Interval-censored semi-competing risks data: a novel approach for modelling bladder cancer. Ph.D. dissertation, Universitat Politècnica de Catalunya, 2010. http://hdl.handle.net/10803/6532

[6] J. M. Satagopan, L. Ben-Porat, M. Berwick, M. Robson, D. Kutler, and A. D. Auerbach. A note on competing risks in survival data analysis. *British Journal of Cancer*, 91(7):1229–1235, 2004.

[7] Meredith S. Shiels, Stephen R. Cole, Joan S. Chmiel, Joseph Margolick, Jeremy Martinson, Zuo-Feng Zhang, and Lisa P. Jacobson. A comparison of ad hoc methods to account for non-cancer AIDS and deaths as competing risks when estimating the effect of HAART on incident cancer AIDS among HIV-infected men. *Journal of Clinical Epidemiology*, 63(4):459–467, 2010.